



Summary

Final-year B. Tech Computer Science student at NMIMS MPSTME with strong programming fundamentals in C++, Java, and Python. Experienced in building end-to-end solutions spanning machine learning pipelines, compiler design, and cryptography. Achieved 78% accuracy and 0.84 ROC AUC on a real-world ML classification task across 3,297 labeled images. Seeking to apply problem-solving and software engineering skills to complex, large-scale engineering challenges.

Education

MPSTME, NMIMS University	2023 - 2027
B. Tech in Computer Science	
CGPA: 3.71	
DAV Public School, Sreshtha Vihar Delhi (Class XII)	2022 - 2023
CBSE: 86.2	
DAV Public School, Sreshtha Vihar Delhi (Class X)	2020 - 2021
CBSE: 91.17	

Projects

SimpLearn Compiler - 2025

Architected a compiler from scratch supporting variable declarations, conditional statements, and loops, applying lexical analysis, parsing, and code execution concepts using core data structures and algorithms.

Github- <https://github.com/Auspicious1/SimpLearn>

Inventory Management System using Operations Research - 2025

Developed a Python-based inventory optimization tool implementing EOQ, Reorder Point, Safety Stock, and Bulk Discount models to reduce holding and ordering costs.

Github- https://github.com/avnigupta01/inventory_mgmt_project

Melanoma Skin Cancer Detection System – 2025

Built a Python-based image processing and machine learning pipeline for skin cancer detection. Extracted ABCD-rule features including asymmetry, border irregularity, color variation, and texture from segmented lesion images. Trained and evaluated Random Forest, SVM, and Logistic Regression on 3,297 labeled images, achieving 78% accuracy and 0.84 ROC AUC on a held-out test set.

Word Jungle – 2026

Engineered an AI-based word ladder game combining heuristic search algorithms with genetic algorithms to solve puzzles and generate valid word-transformation hints in real time.

Day Scholar In-Out Tracking System - 2025

Developed a campus attendance tracking system in Java/HTML/CSS with barcode-scanner integration to record student entry and exit in real time.

SchoolBridge – 2026

Designed a user-centered school communication platform applying HCI and UI/UX principles to improve accessibility, usability, and real-time parent-teacher interaction.

Internships

Maincrafts Technology - June 2026

- Performing data preprocessing and feature engineering on real-world datasets to prepare inputs for machine learning regression models.
- Building and training multiple regression models for predictive analytics using Python and scikit-learn.
- Evaluating and comparing model performance through visualization and metric analysis to identify the most accurate predictive approach.



Certifications

Java Course by UpSkill

Youth Empowerment and Skills (YES+!) Certificate

HackerRank Python (Basic) Certificate – July 2026

HackerRank SQL (Basic) Certificate – July 2026

HackerRank Problem Solving (Basic) Certificate – July 2026

CORE SKILLS

Technical Skills

- **Languages:** C++, Java, Python, R
- **Web:** HTML, CSS, JavaScript
- **Data & ML:** scikit-learn, NumPy, Pandas, SQL
- **Data Visualization:** Tableau, Excel, Power BI
- **Tools & Platforms:** VS Code, Figma, Google Colab, Git
- **Concepts:** Data Structures & Algorithms, OOP, Database Management

Soft Skills

- Willingness to learn
- Cross-functional collaboration
- Multitasking abilities
- Analytical and problem-solving skills
- Teamwork
- Communication Skills

Achievements and Positions of Responsibility

- Participated in LoRRI AI Hackathon organized by LogisticsNow 2026
- Secured 2nd rank in class with a GPA of 3.89 in Semester 5
- Ranked in the Top 50 at Yugantar Inter-Institute Hackathon 2025
- Organizing Committee Member, Documentation – Coding Club 2025
- Organizing Committee Member, Delegate Service & Hospitality – NMMUN 2025
- Organizing Committee Member, Marketing – Saturday 10am Club 2025
- Organizing Committee Member, Creativity – Ambiora Tech Fest 2024
- Participated as a Delegate in NMMUN 2024